

6ELEN018W - Applied Robotics
Lecture 3: Modeling a Robot with Discrete
States and Actions

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The Most General and Challenging Control Problem for Robots

Robots need to operate:

- ▶ In unknown environments (be adaptive and including operation with sensor noise)
- ▶ Cope with high non-linear dynamics when interacting with other systems
- ▶ Be reconfigurable (in the case where part of the robot gets damaged and its dynamics change)

Markov Models

To formulate the general control problem for a robot, Markov models are useful.

A finite state Markov chain (stochastic finite state machine) can be defined:

- ▶ States: $s \in \{1, \dots, m\}$, where m is finite.
- ▶ Starting state s_0 : may be fixed or drawn from some a priori distribution $P_0(s_0)$.
- ▶ Transitions (dynamics): how the system moves from the current state s_t to the next state s_{t+1} .
- ▶ The transitions satisfy the first order Markov property:

$$P(s_{t+1}|s_t, s_{t-1}, \dots, s_0) = P_1(s_{t+1}|s_t) \quad (1)$$

Markov Chains (cont'd)

Markov chains define a stochastic system which generates a sequence of states:

$$s_0 \longrightarrow s_1 \longrightarrow s_2 \longrightarrow \dots$$

where s_0 is drawn from $P_0(s_0)$ and each s_{t+1} from one step transition probabilities $P_1(s_{t+1}|s_t)$.

- ▶ A Markov chain can be represented as a state transition diagram.

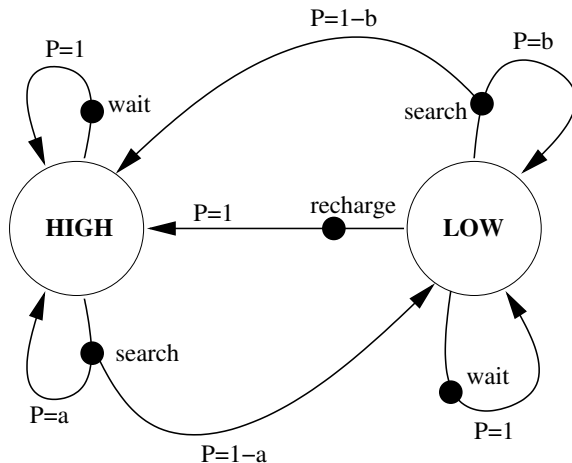
Deterministic vs Stochastic Transition Diagrams

- ▶ Not all systems can consist of transitions that are executed in a deterministic way.

A process whose transitions to different states depend on probabilities is called a *stochastic* process.

Example: A mobile robot has the job of collecting empty cans in an office environment. It has sensors for detecting cans and an arm and gripper to pick them up and place them in an on board bin. It runs on a rechargeable battery. High level decisions about how to search for cans are made by a neural network controller based on the current charge level of the battery. At each point in time, the robot can either 1) actively search for a can for a certain period of time or 2) remain stationary and wait for someone to bring it a can or 3) head back to the recharge station.

The Recycling Robot Example (cont'd)



Transition Probabilities

The conditional probability p_{ij} is defined as the probability that a system which occupies state i , will occupy state j after its next transition.

- ▶ Since the system must be in some state after its next transition:

$$\sum_{j=1}^N p_{ij} = 1 \quad (2)$$

- ▶ Since p_{ij} are probabilities:

$$0 \leq p_{ij} \leq 1 \quad (3)$$

Basic Formulas for Probabilities

- ▶ *Product Rule*: probability $P(A \vee B)$ of a conjunction of two events A and B:

$$P(A \wedge B) = P(A|B)P(B) = P(B|A)P(A)$$

- ▶ *Sum Rule*: probability of a disjunction of two events A and B:

$$P(A \vee B) = P(A) + P(B) - P(A \wedge B)$$

- ▶ *Theorem of total probability*: if events $A_1, \dots, A_n$ are mutually exclusive with $\sum_{i=1}^n P(A_i) = 1$, then

$$P(B) = \sum_{i=1}^n P(B|A_i)P(A_i)$$

Example - The Robot Maker

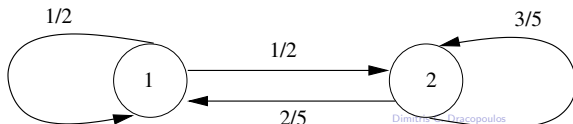
A robot maker is involved in the novelty robot business. He may be in either of two states:

1. The robot he is currently producing has found great favour with the public.
2. The robot is out of favour.

Transition probabilities:

- ▶ If in first state 50% chance of remaining in state 1, and 50% chance of unfortunate move to state 2 at following week.
- ▶ While in state 2, he experiments with new robots and he may return to state 1 after a week with probability $\frac{2}{5}$, or remain unprofitable in state 2 with probability $\frac{3}{5}$.

$$P = [p_{ij}] = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{2}{5} & \frac{3}{5} \end{bmatrix} \quad (4)$$



Markov Chain Problems

- ▶ *Prediction*: Probabilities that the system will be in state s_k after n transitions, given that at $n = 0$ it is in a known state.
- ▶ *Estimation*: Calculation of transition probabilities given some observed sequences of state transitions.

The Prediction Problem

Example: What is the probability that the robot maker will be in state 1 after n weeks, given that he is in state 1 at the beginning of the n -week period?

Define $\pi_i(n)$ as the probability that the system will occupy state i after n transitions, if its state at $n = 0$ is known.

Then:

$$\sum_{i=1}^N \pi_i(n) = 1 \quad (5)$$

$$\pi_j(n+1) = \sum_{i=1}^N \pi_i(n) p_{ij} \quad n = 0, 1, 2, \dots \quad (6)$$

The Prediction Problem (cont'd)

Define a row vector of state probabilities $\pi(n)$ with components $\pi_i(n)$.

Then:

$$\pi(n+1) = \pi(n)P \quad n = 0, 1, 2, \dots \quad (7)$$

Now:

$$\begin{aligned}\pi(1) &= \pi(0)P \\ \pi(2) &= \pi(1)P = \pi(0)P^2 \\ \pi(3) &= \pi(2)P = \pi(0)P^3\end{aligned} \quad (8)$$

In general:

$$\pi(n) = \pi(0)P^n \quad n = 0, 1, 2, \dots \quad (9)$$

Application to the Robot Maker Example

Assume that the robot maker starts with a successful robot, then $\pi_1(0) = 1$, $\pi_2(0) = 0$.

$$\pi(1) = \pi(0)P = [1 \ 0] \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{2}{5} & \frac{3}{5} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

After 1 week the robot maker is equally likely to be successful or unsuccessful.

After 2 weeks:

$$\pi(2) = \pi(1)P = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{2}{5} & \frac{3}{5} \end{bmatrix} = \begin{bmatrix} \frac{9}{20} & \frac{11}{20} \end{bmatrix}$$

so that the robot maker is slightly more likely to be unsuccessful.

Example: Successive State Probabilities Starting with a Successful Robot

n	0	1	2	3	4	5	...
$\pi_1(n)$	1	0.5	0.45	0.445	0.4445	0.44445	...
$\pi_2(n)$	0	0.5	0.55	0.555	0.5555	0.55555	...

As n becomes very large:

- ▶ $\pi_1(n)$ approaches $\frac{4}{9}$
- ▶ $\pi_2(n)$ approaches $\frac{5}{9}$

Ergodic Processes

In the previous example of the toymaker, as n becomes large, $\pi_1(n)$ approaches $\frac{4}{9}$ and $\pi_2(n)$ approaches $\frac{5}{9}$ independent of the starting state.

A Markov process is completely *ergodic* if:

- ▶ The limiting state probability distribution is independent of starting conditions.

For completely ergodic Markov processes π with components π_i is $\pi(n)$ as n approaches infinity. Also called absolute state probabilities.

Absolute State Probabilities

For ergodic Markov processes from (7):

$$\pi = \pi P \quad (10)$$

We also have:

$$\sum_{i=1}^N \pi_i = 1 \quad (11)$$

From (10), (11) the limiting state probabilities for any process can be found.

Ergodic Processes (cont'ed)

For the toymaker example, equations (10), (11) yield:

$$\begin{aligned}\pi_1 &= \frac{1}{2}\pi_1 + \frac{2}{5}\pi_2 \\ \pi_2 &= \frac{1}{2}\pi_1 + \frac{3}{5}\pi_2 \\ \pi_1 + \pi_2 &= 1\end{aligned}\tag{12}$$

The unique solution of the above is: $\pi_1 = \frac{4}{9}$, $\pi_2 = \frac{5}{9}$.

- Finding the limiting state probabilities involves the solution of a set of N linear equations.

Hidden Markov Models

- ▶ A hidden Markov model (HMM) is a model which we generate a sequence of outputs in addition to the Markov state sequence:

$$s_0 \longrightarrow s_1 \longrightarrow s_2 \longrightarrow \dots$$



$$x_0 \quad x_1 \quad x_2$$

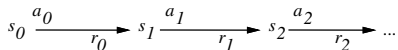
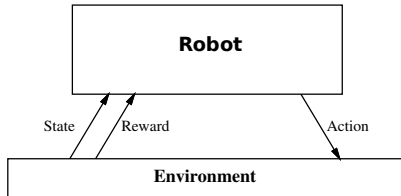
- ▶ Only the outputs $\{x_0, x_1, \dots, x_n\}$ are observed. The state sequence remains hidden.

Hidden Markov Problems to Solve

Two of the most interesting problems to solve, related to HMM are:

- ▶ Calculate the probability that our model generated the observation sequence $\{x_0, x_1, \dots, x_n\}$?
 - ▶ forward-backward algorithm
- ▶ How do we find the most likely hidden state sequence corresponding to these observations?
 - ▶ dynamic programming

The Reinforcement Learning Problem for a Robot



Goal: Learn to choose actions that maximise:

$$r_0 + \gamma r_1 + \gamma^2 r_2 + \dots,$$

where $0 \leq \gamma < 1$

Robot's Learning Task

Execute actions in environment, observe results, and

- ▶ learn action policy $\pi : S \longrightarrow A$ that maximises

$$E[r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \dots]$$

from any starting state in S

- ▶ here $0 \leq \gamma < 1$ is the discount factor for future rewards